Aakash Kumar Singh

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Education

Indian Institute of Science, Bangalore — **CGPA: 8.5/10**
Master of Technology by Research in Computational and Data Sciences — *Aug 2023 – Present*

Maharaja Agrasen Institute of Technology, Delhi — **CPI: 8.94/10**
Bachelor of Technology in Computer Science and Engineering — *Aug 2019 – May 2023*

Publication

Aakash Kumar Singh, Dey P., Shrivatsa S., & Babu R. V. (2025). **Concept Siever: A Method for Concept Forgetting in Text-to-Image Models via Fine-tuning**. Transactions on Machine Learning Research (TMLR).

Aakash Kumar Singh, Joseph K. J., Karanam S., & Babu R. V. (2025). **Forget with Care: A Domain-Agnostic Framework for Concept Erasure without Side-effects** *[Extended Abstract]*. Proceedings of the U&ME Workshop at the IEEE/CVF International Conference on Computer Vision (ICCV).

Birdbath Singhal*, **Aakash Kumar Singh***, Srihari K G, Ankit Dhiman, Venkatesh Babu Radhakrishnan (2025). **Sensitive Gaussians and Where to Find Them: Test Time Adaptation of 3D Gaussians**. *[Under Review]*.

Projects

Concept Forgetting for Text to Image Diffusion Model — **May 2024-August 2025**
Prof. Venkatesh Babu R.
- Developed **Concept Siever**, a novel domain-agnostic framework for concept erasure in diffusion models with minimal side-effects, leading to a **TMLR publication.**
- Demonstrated **state-of-the-art** performance by improving **concept forgetting efficacy by over** 33% **on the I2P benchmark** ; this superior result was confirmed in a user study where **our method was preferred** 72.8% **of the time**.
- Introduced the **first framework to offer fine-grained, inference-time concept-level control over forgetting strength** , allowing for dynamic adjustment without any retraining

Enhancing Video Diffusion with Parametric Human Models — **July 2025-Present**
Prof. Venkatesh Babu R.
- Researching **realistic human motion** generation within **video diffusion models** across generation, inpainting, and frame interpolation.
- Developing a novel method to enhance the realism of generated videos by **integrating the SMPL parametric human body model into the diffusion process**.
- Currently developing a **training-free method for the targeted editing of human subjects in video**, enabling precise control over their motion, location, and speed.

Research Experience

Project Associate — **September 2024-Present**
Vision and AI Lab (VAL)
- Developed a **zero-shot motion personalisation** method for **any-to-video generative models**, enabling **realistic human motion generation** and achieving appreciable improvement in motion fidelity.
- Investigating the internal mechanisms of generative video diffusion models by applying **mechanistic interpretability** methods to map and influence the model's content generation pathways.
- Collaborated with global PhD scholars and research scientists from Adobe and Flipkart to develop novel methods for **controllable video generation.**

Achievements

Oral presentation at ICCV workshop — **2025**
*Invited to deliver a 10-minute presentation on our extended abstract **"Forget with Care"** recognised by Unlearning and Model Editing in Vision Workshop at ICCV 2025.*

Vision India Track — **2025**
***Concept Siever** was accepted for presentation in the Vision India Track at the Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP) 2025.*

Ministry of Education Post Graduate Scholarship — **2023**
*Secured a **national merit-based fellowship** from the **Ministry of Education for postgraduate studies**, awarded for exceptional performance in the Graduate Aptitude Test in Engineering (GATE).*

Graduate Aptitude Test in Engineering (GATE) — **2023**
*Secured **All India Rank 219 (Top 0.3%)** in GATE CS 2023*

Academic Service

Teaching Assistant **Indian Institute of Science, Bangalore**

- DS 265o Deep Learning for Visual Analytics (August 2024 Term & August 2025 Term)
- DS 265 Deep Learning for Computer Vision (January 2024 Term)

Peer Review **2024 - Present**

- Transactions on Machine Learning Research (TMLR)
- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- IEEE Transactions on Image Processing (TIP)
- IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)

Research Mentor & Team Lead **Vision and AI Lab, Indian Institute of Science**

- Guided mentees through the **full research lifecycle**, from literature review and ideation to implementation and manuscript preparation, fostering their growth into independent contributors.
- Led and mentored a cohort of 4 junior researchers (Project Associates, interns, and undergraduates), overseeing their integration into high-level Generative AI projects.